

Software Requirements Specification (SRS)

Project: Smart Complaint Management System (SCMS)

Version: 1.0

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1.0	05-09-2026	Team 9	Initial SRS draft for Smart Complaint Management System	Pending

Approvals

Role	Name	Signature / Email	Date
Team Lead			

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Smart Complaint Management System (SCMS), a web application that allows citizens to lodge, track, and receive resolutions for civic/organizational complaints, and allows administrators to manage, assign, and resolve those complaints. It defines the functional and non-functional requirements, interfaces, and verification criteria for the system, intended for developers, evaluators, and course instructors.

1.2 Scope

SCMS covers citizen-facing complaint lodging and tracking, an administrative dashboard for triage, assignment, and resolution, automated notifications, and basic analytics/reporting. It excludes payment processing, third-party ticketing system integration, and native mobile applications.

1.3 Audience

Developers, QA/Testers, Course Instructors/Evaluators, System Administrators, and End Users (Citizens).

1.4 Definitions, Acronyms, and Abbreviations

- SCMS – Smart Complaint Management System
- OTP – One-Time Password
- UI – User Interface
- API – Application Programming Interface
- RBAC – Role-Based Access Control
- SLA – Service Level Agreement
- JWT – JSON Web Token
- ORM – Object-Relational Mapping
- MVC – Model-View-Controller
- CSRF – Cross-Site Request Forgery
- XSS – Cross-Site Scripting
- VCS – Version Control System
- CI/CD – Continuous Integration / Continuous Deployment
- RTM – Requirements Traceability Matrix

2. Overall description

2.1 Product perspective

SCMS is a new, standalone web-based system consisting of a citizen-facing portal, an administrative dashboard, a backend REST API, and a relational database. It follows a client-server architecture with a browser-based frontend and a backend built using Python (Flask/Django) or PHP (Laravel), interacting through well-defined REST APIs. Notification services (email/SMS) are integrated as external dependencies.

2.2 Major product functions

- Citizen registration and authentication
- Complaint lodging with category, description, location, and photo evidence
- Unique complaint ID generation and status tracking
- Admin dashboard for triage, categorization, and assignment
- Complaint resolution workflow with status updates and remarks
- SLA-based auto-escalation of overdue complaints
- Email/SMS/in-app notifications on status change
- Citizen feedback and rating on resolution
- Analytics and reporting (category-wise, department-wise, turnaround time)

2.3 User roles and characteristics

- Complainant (Citizen): general public with basic digital literacy; expects a simple form to lodge complaints and clear status visibility.
- Admin/Officer: departmental staff responsible for reviewing, assigning, and resolving complaints; needs a dashboard with filters and bulk actions.

2.4 Operating environment

Server-side: Linux-based hosting environment running Python (Flask/Django) or PHP (Laravel) with a relational database (PostgreSQL/MySQL). Client-side: modern web browsers (Chrome, Firefox, Edge, Safari) on desktop and mobile. Deployed behind HTTPS with a reverse proxy (e.g., Nginx).

2.5 Constraints

- Must be delivered using Python (Flask/Django) or PHP (Laravel) as per project specification.
- Must run on standard, low-cost hosting suitable for an academic project (no proprietary infrastructure).
- Database schema must support audit logging and traceability for evaluation.
- Development timeline is constrained to a single academic semester.
- Source code shall be version-controlled on GitHub and all requirements/tasks tracked on Jira for the duration of the project.

3. External interface requirements

3.1 User interfaces

A responsive web UI with a simple complaint-lodging form for citizens, a searchable/filterable dashboard for admins, and a status-timeline view for tracking. Accessible design with readable fonts and color-contrast compliant status indicators.

3.2 Hardware interfaces

None beyond standard client devices (desktop/laptop/smartphone) with a camera (for photo evidence capture) and internet connectivity. No specialized hardware is required.

3.3 Software interfaces

- Backend REST API (JSON over HTTPS) for all client-server communication.
- Relational database (PostgreSQL/MySQL) accessed via ORM (SQLAlchemy/Django ORM or Eloquent).
- Email/SMS gateway API (e.g., SMTP, Twilio, or equivalent) for notifications and OTP delivery.
- Cloud/local file storage service for complaint photo attachments.

3.4 Communications

- HTTPS/TLS 1.2+ enforced for all client-server traffic.
- REST API with JSON payloads; JWT/session-based authentication tokens for authenticated requests.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow SCMS-F-####.

4.1 User Registration & Authentication

Description: Allow citizens and administrators to securely register and authenticate before accessing role-specific functionality.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SCMS-F-001	The system shall allow a citizen to register using name, email/mobile number, and password, with email/OTP verification.	Functional	High	Complainant	AC-SCMS-F-001: Account created only after OTP/email verification. Test: TC-Reg-01	Requires SMTP/OTP gateway

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SCMS-F-002	The system shall authenticate registered users via email/mobile and password before granting access to complaint features.	Functional	High	Complainant / Admin	AC-SCMS-F-002: Invalid credentials rejected with generic error. Test: TC-Auth-01	JWT/session based auth
SCMS-F-003	The system shall hash and salt all stored passwords and never store or log plaintext passwords.	Functional	High	Security	AC-SCMS-F-003: DB inspection shows only hashed values. Test: TC-Sec-01	bcrypt/argon2
SCMS-F-004	The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts and log the event.	Functional	Medium	Security	AC-SCMS-F-004: 6th attempt is blocked and audit entry created. Test: TC-Auth-02	Rate-limiting middleware

4.2 Complaint Lodging

Description: Allow authenticated citizens to submit a new complaint with all details required for triage and resolution.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
SCMS-F-010	The system shall allow an authenticated citizen to lodge a new complaint with category, description, location, and optional photo evidence.	Functional	High	Complainant	AC-SCMS-F-010: Complaint saved with a unique Complaint ID and status 'Submitted'. Test: TC-Lodge-01	File upload module
SCMS-F-011	The system shall auto-generate a unique, trackable Complaint ID and display it to the user immediately after submission.	Functional	High	Complainant	AC-SCMS-F-011: Complaint ID follows format SCMS-YYYY-NNNNN and is unique. Test: TC-Lodge-02	ID generation service
SCMS-F-012	The system shall validate mandatory fields (category, description, location) and reject incomplete submissions with clear error messages.	Functional	High	Complainant	AC-SCMS-F-012: Submission blocked until all mandatory fields are filled. Test: TC-Lodge-03	Client + server-side validation
SCMS-F-013	The system shall allow attachment of up to 3 images (max 5 MB each) as supporting evidence for a complaint.	Functional	Medium	Complainant	AC-SCMS-F-013: Files beyond limit/size are rejected with a message. Test: TC-Lodge-04	File storage (local/cloud)

4.3 Complaint Tracking & Status Updates

Requirements: The system shall allow complainants to search/track a complaint using Complaint ID or account and view a status timeline (Submitted, In Review, Assigned, In Progress, Resolved, Closed, Reopened). The system shall notify complainants of status changes through in-app notifications and, where configured, email/SMS. IDs: SCMS-F-020 to SCMS-F-024.

4.4 Admin Resolution & Assignment

Requirements: The system shall let Admins view all complaints in a dashboard, filter/sort by category, priority, and status, assign complaints to the appropriate department/officer, update status with resolution remarks, and escalate overdue complaints automatically based on SLA. IDs: SCMS-F-030 to SCMS-F-035.

4.5 Notifications, Feedback & Reporting

Requirements: The system shall notify complainants of status changes through in-app notifications and, where configured, email/SMS, allow citizens to rate resolution quality and leave feedback, and allow Admins to generate category-wise and department-wise resolution reports. IDs: SCMS-F-040 to SCMS-F-042.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs SCMS-NF-####.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
SCMS-NF-001	The system shall load the complaint dashboard and return search results within 3 seconds for 90% of requests under normal load (up to 500 concurrent users).	Performance	High	90th percentile $\leq$ 3s in load test. Test: TC-Perf-01
SCMS-NF-002	The system shall maintain 99.5% uptime monthly, excluding scheduled maintenance windows.	Reliability/Availability	High	Uptime monitoring reports $\geq$ 99.5% per month. Test: Ops reports
SCMS-NF-003	All personally identifiable information (name, contact, address) shall be encrypted at rest and in transit.	Security/Compliance	High	Security audit / DB inspection confirms encryption. Test: TC-Sec-02
SCMS-NF-004	The web interface shall be responsive and usable on screen widths from 360px (mobile) to 1920px (desktop).	Usability	Medium	Manual/automated responsive test across breakpoints. Test: TC-UX-01
SCMS-NF-005	The system shall be horizontally scalable to support growth from 1,000 to 50,000 registered users without architectural change.	Scalability	Medium	Load test at 10x baseline shows acceptable degradation. Test: TC-Scale-01
SCMS-NF-006	The codebase shall follow a modular MVC structure (Flask/Django or Laravel) enabling a new module to be added within 3 developer-days.	Maintainability	Medium	Code review checklist; module addition time-boxed test. Test: TC-Maint-01

5.1 Security

5.1.1 Security Objectives

- Confidentiality: Protect citizen personal data (name, contact, address, uploaded evidence) from unauthorized access or disclosure.
- Integrity: Ensure complaint records, status history, and audit logs cannot be altered without detection.
- Availability & Accountability: Ensure the system remains available to citizens and admins, and that every state-changing action is attributable to an authenticated user.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
SCMS-SR-001	TLS 1.2+ shall be mandatory for all client-server and server-database network connections.	Security	High	AC-SR-001: Traffic capture shows only TLS 1.2+. Test: TC-Sec-03

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
SCMS-SR-002	The system shall implement role-based access control (RBAC) restricting Admin/Officer functions from Complainant accounts.	Security	High	AC-SR-002: Complainant account cannot access admin routes. Test: TC-Sec-04
SCMS-SR-003	The system shall sanitize and validate all user inputs to prevent SQL injection and Cross-Site Scripting (XSS).	Security	High	AC-SR-003: OWASP ZAP scan shows no critical injection/XSS findings. Test: TC-Sec-05
SCMS-SR-004	The system shall maintain a tamper-evident audit log of all status changes, assignments, and admin actions.	Security	Medium	AC-SR-004: Audit log entries are immutable and timestamped. Test: TC-Sec-06
SCMS-SR-005	The system shall implement CSRF protection tokens on all state-changing forms and API endpoints.	Security	High	AC-SR-005: CSRF token missing/invalid requests are rejected. Test: TC-Sec-07

6. Quality attributes & Acceptance tests

Exit criteria for acceptance: All High-priority functional requirements implemented and verified, no critical NFR or security-requirement failures, and the RTM shows all listed test cases in status 'Passed' (P).

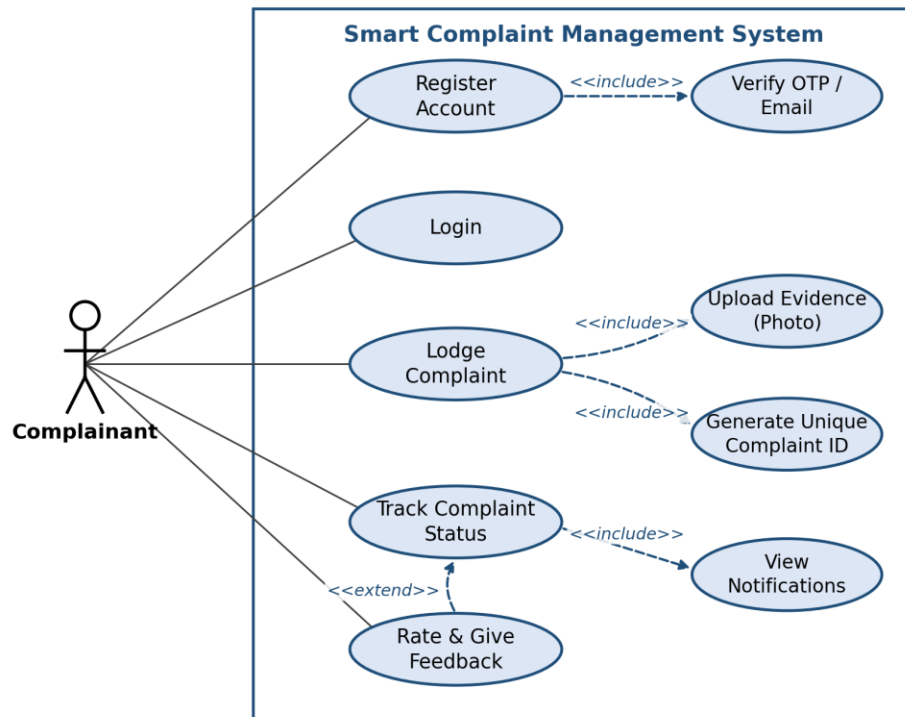
Acceptance test suites: Registration & Authentication, Complaint Lodging, Tracking, Admin Assignment & Resolution, Notifications, Performance, Security, and Usability/Accessibility tests.

7. System models and diagrams

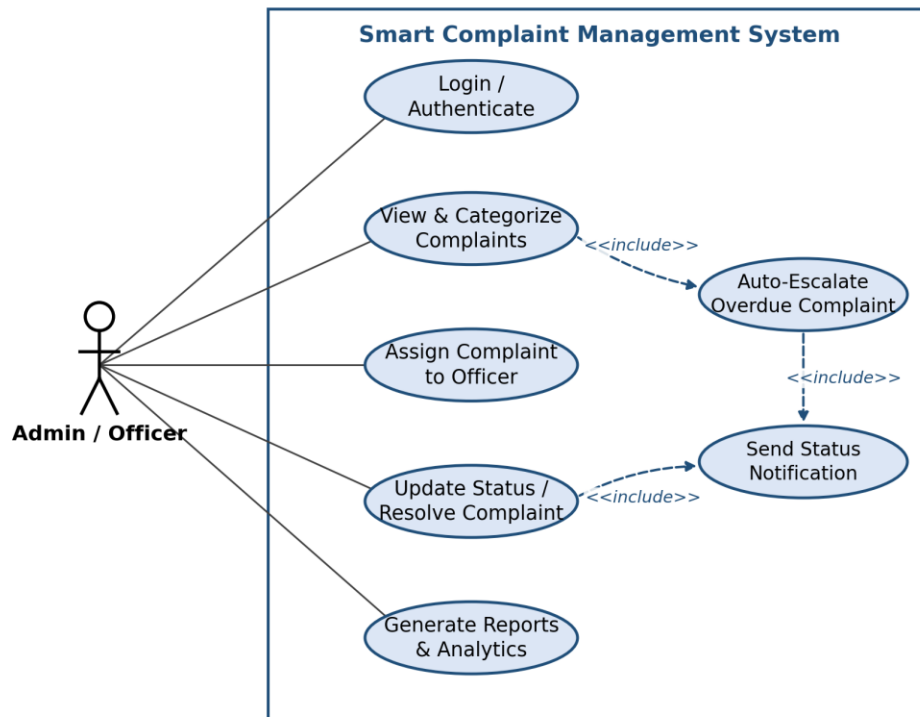
7.1 UML Use-Case Diagrams

The following two use-case diagrams illustrate the primary interactions of the Complainant and the Admin/Officer with the Smart Complaint Management System.

Use-Case Diagram 1: Complainant Interactions with SCMS



Use-Case Diagram 2: Admin/Officer Interactions with SCMS



8. Requirements Traceability Matrix (RTM)

Requirements are traced across the SDLC using the Requirements Traceability Matrix below. Status column uses N = Not tested, P = Passed, A = tested and Approved.

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
SCMS-F-001	Citizen registration with OTP	4.1 / DS-Auth-01	AuthModule	TC-Reg-01	N	
SCMS-F-002	Authenticate registered user	4.1 / DS-Auth-02	AuthModule	TC-Auth-01	N	
SCMS-F-003	Hash and salt stored passwords	4.1 / DS-Auth-03	AuthModule	TC-Sec-01	N	
SCMS-F-004	Lock account after failed logins	4.1 / DS-Auth-04	AuthModule	TC-Auth-02	N	
SCMS-F-010	Lodge new complaint	4.2 / DS-Complaint-01	ComplaintModule	TC-Lodge-01	N	
SCMS-F-011	Auto-generate Complaint ID	4.2 / DS-Complaint-02	ComplaintModule	TC-Lodge-02	N	
SCMS-F-012	Validate mandatory complaint fields	4.2 / DS-Complaint-03	ComplaintModule	TC-Lodge-03	N	
SCMS-F-013	Attach photo evidence to complaint	4.2 / DS-Complaint-04	ComplaintModule	TC-Lodge-04	N	
SCMS-F-020	Track complaint status by ID	4.3 / DS-Track-01	TrackingModule	TC-Track-01	N	
SCMS-F-021	Track complaint status by account	4.3 / DS-Track-02	TrackingModule	TC-Track-02	N	
SCMS-F-022	View complaint status timeline	4.3 / DS-Track-03	TrackingModule	TC-Track-03	N	
SCMS-F-023	In-app / email-SMS status notifications	4.3 / DS-Track-04	NotificationModule	TC-Track-04	N	
SCMS-F-024	Reopen resolved or closed complaint	4.3 / DS-Track-05	TrackingModule	TC-Track-05	N	
SCMS-F-030	Assign complaint to officer	4.4 / DS-Admin-01	AdminModule	TC-Admin-01	N	
SCMS-F-031	View all complaints in admin dashboard	4.4 / DS-Admin-02	AdminModule	TC-Admin-02	N	
SCMS-F-032	Filter/sort complaints by category, priority, status	4.4 / DS-Admin-03	AdminModule	TC-Admin-03	N	
SCMS-F-033	Update status with resolution remarks	4.4 / DS-Admin-04	AdminModule	TC-Admin-04	N	
SCMS-F-034	Auto-escalate overdue complaints (SLA)	4.4 / DS-Admin-05	AdminModule	TC-Admin-05	N	
SCMS-F-035	Reassign complaint to different officer/dept.	4.4 / DS-Admin-06	AdminModule	TC-Admin-06	N	
SCMS-F-040	Send status-change notification	4.5 / DS-Notify-01	NotificationModule	TC-Notify-01	N	

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
SCMS-F-041	Rate resolution & give feedback	4.5 / DS-Feedback-01	FeedbackModule	TC-Feedback-01	N	
SCMS-F-042	Generate category/department-wise reports	4.5 / DS-Report-01	ReportingModule	TC-Report-01	N	
SCMS-NF-001	Dashboard response time target	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N	
SCMS-NF-002	99.5% monthly uptime	5 / DS-Reliability-01	Monitoring	Ops reports	N	
SCMS-NF-003	Encryption of PII	5 / DS-Sec-01	DataLayer	TC-Sec-02	N	
SCMS-NF-004	Responsive UI (360px-1920px)	5 / DS-UX-01	WebUI	TC-UX-01	N	
SCMS-NF-005	Horizontal scalability (1k-50k users)	5 / DS-Scale-01	Infrastructure	TC-Scale-01	N	
SCMS-NF-006	Modular MVC codebase	5 / DS-Maint-01	Codebase	TC-Maint-01	N	
SCMS-SR-001	TLS 1.2+ mandatory on all connections	5.1.2 / DS-Sec-03	Network	TC-Sec-03	N	
SCMS-SR-002	Role-based access control	5.1.2 / DS-Sec-02	AuthModule	TC-Sec-04	N	
SCMS-SR-003	Input sanitization (SQLi / XSS)	5.1.2 / DS-Sec-04	InputValidation	TC-Sec-05	N	
SCMS-SR-004	Tamper-evident audit log	5.1.2 / DS-Sec-05	AuditLogModule	TC-Sec-06	N	
SCMS-SR-005	CSRF protection on state-changing requests	5.1.2 / DS-Sec-06	WebUI / API	TC-Sec-07	N	